

Reflection Journal

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Learning Reflection

This lab helped me understand that machine learning depends on several tools working together rather than on one library alone. Before working with the Iris dataset, I viewed pandas, NumPy, and matplotlib mostly as separate Python libraries. Working through the notebook made their roles clearer. Pandas was useful for organizing the dataset into a DataFrame and grouping information by species, while NumPy made numerical calculations such as the mean and standard deviation straightforward. Matplotlib added another way of understanding the information because I could see patterns instead of relying only on printed numbers.

One useful insight came from comparing the numerical results with the scatter plot. The Iris dataset contains 150 samples, four features, and three classes. The visualization made it easier to notice that the setosa observations are more clearly separated from the other species, while versicolor and virginica have more overlap. This showed me why visualization is important in data analysis: a table of values can provide exact information, but a graph can reveal patterns and relationships much more quickly.

I also became more comfortable with the idea of using different tools for different parts of a machine learning workflow. The DataFrame made the data easier to inspect and analyze, NumPy handled numerical operations, and matplotlib turned the results into a visual explanation. The basic statistical analysis also showed me that averages can summarize a dataset, but they do not tell the entire story. Looking at the distribution of the observations is important because two groups can have similar summary statistics while still showing different patterns.

Another important learning point was working with code that already contained a structure and verification steps. The notebook used printed results and checks to confirm that calculations produced the expected types and outputs. Seeing a message such as “Task 1 completed successfully” helped me

understand the value of testing code instead of assuming that it worked. This also made error checking feel less intimidating because each result could be examined before moving to the next task.

The GitHub section connected the technical work to a real project workflow. I learned that version control is not only about storing code; it also provides a way to organize project files, track changes, collaborate, and keep a record of work. The idea of committing changes as snapshots was especially useful because it gives a project a history instead of leaving only one final version.

Overall, this lab changed how I think about the beginning of a machine learning project. I now see data preparation, numerical analysis, visualization, documentation, and version control as connected parts of the same workflow. One question I would like to explore further is which Iris features are most useful for predicting the species and how accurately a machine learning classifier could separate the three classes. That question connects the exploratory work in this lab to the modeling work that comes next.